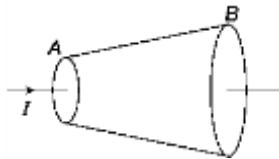


PHYSICS

1. **Assertion:** Current I is flowing through a cylindrical wire of non-uniform cross-section as shown. Section of wire near A will be more heated compared to the section near B.



Reason: Current density near A is more.

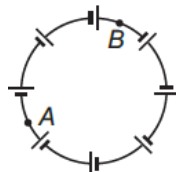
- (1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
 (2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
 (3) If Assertion is true, but the Reason is false.
 (4) If Assertion is false but the Reason is true.
2. **Assertion :** In the part of a circuit shown in figure, given that $V_b > V_a$. The current should flow from b to a.



Reason : Direction of current inside a battery is always from negative terminal to positive terminal.

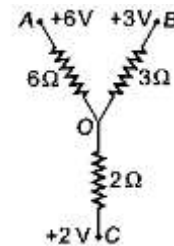
- (1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
 (2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
 (3) If Assertion is true, but the Reason is false.
 (4) If Assertion is false but the Reason is true.
3. The unit of electrical conductivity is –
 (1) ohm-m^{-2} (2) $\text{ohm} \times \text{m}$
 (3) $\text{ohm}^{-1}\text{-m}^{-1}$ (4) None of these
4. If the length of the filament of a heater is reduced by 10%, the power of the heater will
 (1) increase by about 9%
 (2) increase by about 11%
 (3) increase by about 19%
 (4) decrease by about 10%

5. N identical current sources each of emf E and internal resistance r are connected to form a closed loop as shown in figure. The potential difference between points A and B which divides the circuit into n and $(N - n)$ units is

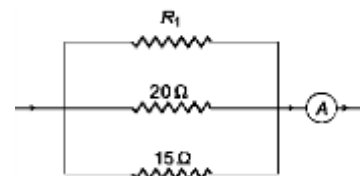


- (1) nE (2) $(N - n)E$
 (3) nE (4) zero

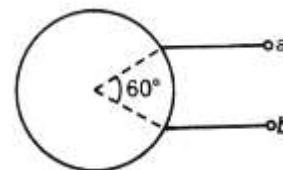
6. Three resistances are joined together to form a letter Y, as shown in figure. If the potentials of the terminals A, B and C are 6V, 3V and 2V respectively, then the potential of the point O will be –



- (1) 4 V (2) 3 V
 (3) 2.5 V (4) 0 V
7. The drift velocity of free electrons in a conductor is v , when a current i is flowing in it. If both the radius and current are doubled, then the drift velocity will be –
 (1) v (2) $v/2$
 (3) $v/4$ (4) $v/8$
8. In the given circuit the current flowing through the resistance $20\ \Omega$ is 0.3 A, while the ammeter reads 0.8 A. What is the value of R_1 ?

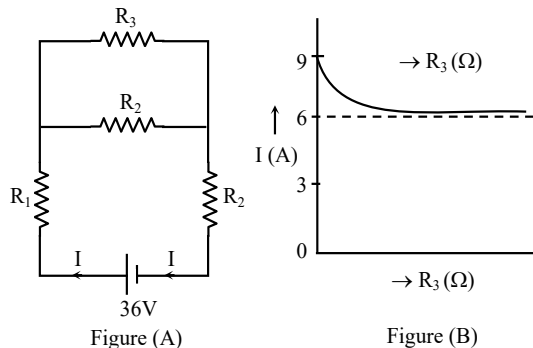


- (1) 30 Ω (2) 40 Ω
 (3) 50 Ω (4) 60 Ω
9. A flow of 10^7 electrons per second in a conducting wire constitutes a current of –
 (1) 1.6×10^{-12} A (2) 1.6×10^{26} A
 (3) 1.6×10^{-26} A (4) 1.6×10^{12} A
10. A uniform wire of resistance $18\ \Omega$ is bent in the form of a circle. The effective resistance across the points a and b is –



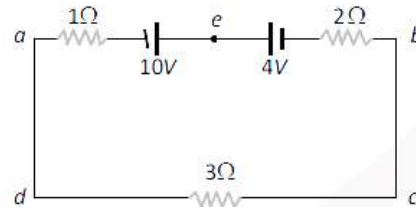
- (1) 3 Ω (2) 2 Ω
 (3) 2.5 Ω (4) 6 Ω

11. In the circuit shown in figure. (a), R_3 is a variable resistance. As the value R_3 is changed, current I through the cell varies as shown. Obviously, the variation is asymptotic, i.e. $I \rightarrow 6\text{A}$ as $R_3 \rightarrow \infty$. Resistances R_1 and R_2 are, respectively –



- (1) $4\ \Omega$, $2\ \Omega$ (2) $2\ \Omega$, $4\ \Omega$
 (2) $2\ \Omega$, $2\ \Omega$ (4) $1\ \Omega$, $4\ \Omega$
12. Find the resistor in which the dissipated power is the greatest
-
- (1) $6\ \Omega$ (2) $9\ \Omega$ (3) $18\ \Omega$ (4) $5\ \Omega$
13. The resistance of a wire is 20 ohms. It is so stretched that the length becomes three times, then the new resistance of the wire will be
- (1) 6.67 ohms (2) 60.0 ohms
 (3) 120 ohms (4) 180.0 ohms
14. Resistivity of iron is 1×10^{-7} ohm-metre. The resistance of the given wire of a particular thickness and length is $1\ \Omega$. If the diameter and length of the wire both are doubled, the resistivity will be -
- (1) 1×10^{-7} (2) 2×10^{-7}
 (3) 4×10^{-7} (4) None of these
15. In a conductor 4 coulombs of charge flows for 2 seconds. The value of electric current will be
- (1) 4 volts (2) 4 amperes
 (3) 2 amperes (4) 2 volts
16. In hydrogen atom, the electron makes 6.6×10^{15} revolutions per second around the nucleus in an orbit of radius $0.5 \times 10^{-10}\text{m}$. It is equivalent to a current nearly
- (1) 1 A (2) 1 mA
 (3) $1\ \mu\text{A}$ (4) $1.6 \times 10^{-19}\text{A}$

17. The magnitude and direction of the current in the circuit shown will be –



- (1) $\frac{7}{3}$ A from a to b through e
 (2) $\frac{7}{3}$ A from b to a through e
 (3) 1A from b to a through e
 (4) 1A from a to b through e
18. A cell of e.m.f. E is connected with an external resistance R , then p.d. across cell is V . The internal resistance of cell will be –
- (1) $\frac{(E - V)R}{E}$ (2) $\frac{(E - V)R}{V}$
 (3) $\frac{(V - E)R}{V}$ (4) $\frac{(V - E)R}{E}$
19. In the circuit shown, potential difference between X and Y will be –
-
- (1) 0 (2) 20 V (3) 60 V (4) 120 V
20. In the circuit shown, A and V are ideal ammeter and voltmeter respectively. Reading of the voltmeter will be
-
- (1) 2 V (2) 1 V
 (3) 0.5 V (4) 0
21. A voltmeter has a resistance of G ohms and range V volts. The value of resistance used in series to convert it into a voltmeter of range nV volts is –
- (1) nG (2) $(n - 1)G$
 (3) $\frac{G}{n}$ (4) $\frac{G}{(n - 1)}$

22. In Wheatstone's bridge $P = 9 \text{ ohm}$, $Q = 11 \text{ ohm}$, $R = 40 \text{ ohm}$ and $S = 6 \text{ ohm}$. How much resistance must be put in parallel to the resistance S to balance the bridge –

- (1) 24 ohm (2) $\frac{44}{9} \text{ ohm}$
(3) 26.4 ohm (4) 18.7 ohm

23. If a long hollow copper pipe carries a direct current, the magnetic field associated with the current will be –

- (1) Only inside the pipe
(2) Only outside the pipe
(3) Neither inside nor outside the pipe
(4) Both inside and outside the pipe

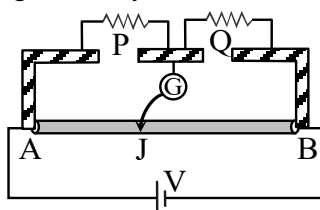
24. The magnetic field $d\vec{B}$ due to a small current element $d\vec{l}$ at a distance $\vec{r}$ and element carrying current i is, or Vector form of Biot-savart's law is

- (1) $d\vec{B} = \frac{\mu_0 i}{4\pi r} \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$
(2) $d\vec{B} = \frac{\mu_0 i}{4\pi r} \left(\frac{d\vec{l} \times \vec{r}}{r^2} \right)$
(3) $d\vec{B} = \frac{\mu_0 i}{4\pi r} \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$
(4) $d\vec{B} = \frac{\mu_0 i}{4\pi r} \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$

25. The magnetic induction at a point P which is distant 4 cm from a long current carrying wire is 10^{-8} Tesla . The field of induction at a distance 12 cm from the same current would be

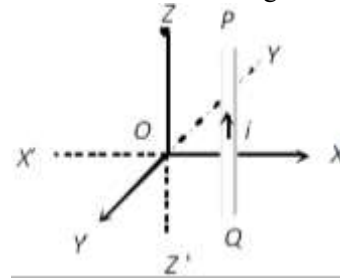
- (1) $3.33 \times 10^{-9} \text{ Tesla}$ (2) $1.11 \times 10^{-4} \text{ Tesla}$
(3) $3 \times 10^{-3} \text{ Tesla}$ (4) $9 \times 10^{-2} \text{ Tesla}$

26. In a meter bridge circuit as shown in figure, the bridge is balanced when $AJ = 20 \text{ cm}$. On interchanging P and Q the balance length shifts by –



- (1) Zero (2) 80 cm
(3) 40 cm (4) 60 cm

27. A vertical wire kept in $Z-X$ plane carries a current from Q to P (see figure). The magnetic field due to current will have the direction at the origin O along –



- (1) OX (2) OX'
(3) OY (4) OY'

28. A charge q coulomb moves in a circle at n revolutions per second and the radius of the circle is r metre. Then magnetic field at the centre of the circle is –

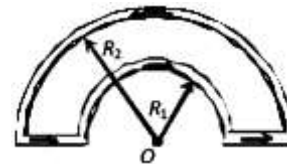
- (1) $\frac{2\pi q}{nr} \times 10^{-7} \text{ N/(amp-meter)}$
(2) $\frac{2\pi q}{r} \times 10^{-7} \text{ N/(amp-meter)}$
(3) $\frac{2\pi nq}{r} \times 10^{-7} \text{ N/(amp-meter)}$
(4) $\frac{2\pi q}{r} \text{ N/(amp-meter)}$

29. An infinitely long straight conductor is bent into the shape as shown in the figure. It carries a current of i ampere and the radius of the circular loop is r metre. Then the magnetic induction at its centre will be –



- (1) $\frac{\mu_0 i}{4\pi r} (\pi + 1)$ (2) $\frac{\mu_0 i}{4\pi r} (\pi - 1)$
(3) 0 (4) ∞

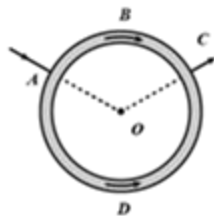
30. The magnetic induction at the centre O in the figure shown is –



- (1) $\frac{\mu_0 i}{4} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ (2) $\frac{\mu_0 i}{4} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$
(3) $\frac{\mu_0 i}{4} (R_1 - R_2)$ (4) $\frac{\mu_0 i}{4} (R_1 + R_2)$

31. A uniform wire is bent in the form of a circle of radius R . A current I enters at A and leaves at C as shown in the figure :

If the length ABC is half of the length ADC , the magnetic field at the centre O will be –

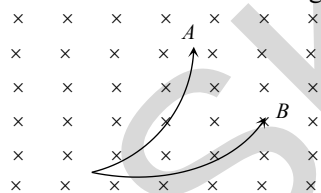


- (1) 0 (2) $\frac{\mu_0 I}{2R}$ (3) $\frac{\mu_0 I}{4R}$ (4) $\frac{\mu_0 I}{6R}$

32. The magnetic induction due to an infinitely long straight wire carrying a current i at a distance r from wire is given by –

- (1) $|B| = \left(\frac{\mu_0}{4\pi} \right) \frac{2i}{r}$
 (2) $|B| = \left(\frac{\mu_0}{4\pi} \right) \frac{r}{2i}$
 (3) $|B| = \left(\frac{4\pi}{\mu_0} \right) \frac{2i}{r}$
 (4) $|B| = \left(\frac{4\pi}{\mu_0} \right) \frac{r}{2i}$

33. The magnetic field is downward perpendicular to the plane of the paper and a few charged particles are projected in it. Which of the following is true

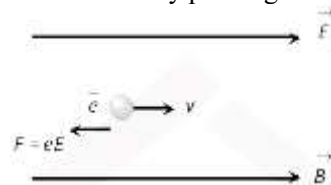


- (1) A represents proton and B and electron
 (2) Both A and B represent protons but velocity of A is more than that of B
 (3) Both A and B represent protons but velocity of B is more than that of A
 (4) Both A and B represent electrons, but velocity of B is more than that of A

34. A long solenoid has a radius a and number of turns per unit length is n . If it carries a current i , then the magnetic field on its axis is directly proportional to

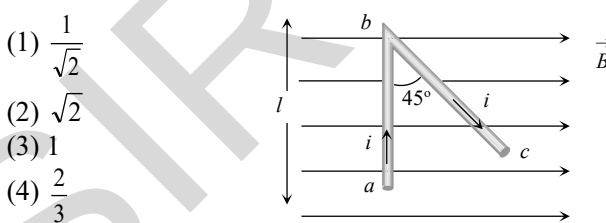
- (1) ani (2) ni (3) $\frac{ni}{a}$ (4) n^2i

35. A uniform electric field and a uniform magnetic field are produced, pointed in the same direction. An electron is projected with its velocity pointing in the same direction –



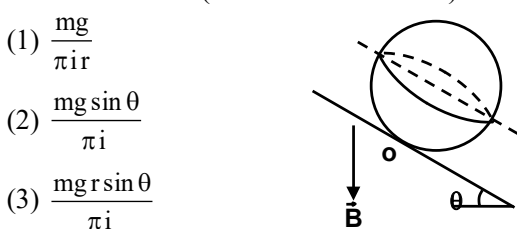
- (1) The electron will turn to its right
 (2) The electron will turn to its left
 (3) The electron velocity will increase in magnitude
 (4) The electron velocity will decrease in magnitude

36. A wire abc is carrying current i . It is bent as shown in fig and is placed in a uniform magnetic field of magnetic induction B . Length $ab = l$ and $\angle abc = 45^\circ$. The ratio of force on ab and on bc is



- (1) $\frac{1}{\sqrt{2}}$
 (2) $\sqrt{2}$
 (3) 1
 (4) $\frac{2}{3}$

37. In the figure shown a coil of single turn is wound on a sphere of radius r and mass m . The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. If sphere is in rotational equilibrium the value of B is (current in the coil is i)



- (1) $\frac{mg}{\pi i r}$
 (2) $\frac{mg \sin \theta}{\pi i}$
 (3) $\frac{mg r \sin \theta}{\pi i}$
 (4) None of these

38. **STATEMENT-1:** A wire of uniform cross section and uniform resistivity is connected across an ideal cell. Now the length of the wire is doubled keeping volume of wire constant. The drift velocity of electrons after stretching the wire becomes one fourth of what it was before stretching the wire.

STATEMENT-2: If a wire (of uniform resistivity and uniform cross-section) of length ℓ_0 is stretched to length $n\ell_0$, then its resistance becomes n^2 times of what it was before stretching the wire (the volume of wire is kept constant in stretching process). Further at constant potential difference, current is inversely proportional to resistance. Finally drift velocity of free electron is directly proportional to current and inversely proportional to cross section area of current carrying wire.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (3) Statement-1 is True, Statement-2 is False
 (4) Statement-1 is False, Statement-2 is True

39. **STATEMENT-1:** Magnitude of potential difference across the terminals of a non-ideal battery in a circuit cannot be greater than its emf.

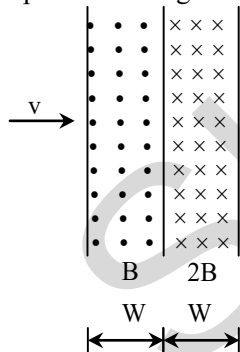
STATEMENT-2: When a current of magnitude I is passing through a battery of emf E and internal resistance r as shown, the magnitude of potential difference (V) across the battery is given by

$$V = E - I r$$



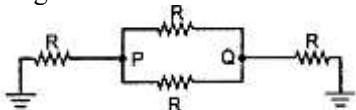
- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (3) Statement-1 is True, Statement-2 is False
 (4) Statement-1 is False, Statement-2 is True.

40. Region of the magnetic field shown in the figure consist of two uniform magnetic fields. If v is the velocity just required for a charge q of particle of mass m and charge to pass through the region of magnetic field. Particle is projected with velocity " v " then how much time does such a charge spend in the region of magnetic field ?



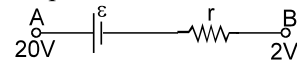
- (1) $\frac{\pi m}{2qB}$ (2) $\frac{\pi m}{qB}$ (3) $\frac{\pi m}{4qB}$ (4) $\frac{3\pi m}{2qB}$

41. The net resistance between points P and Q in the circuit shown in fig. is –



- (1) $\frac{R}{2}$ (2) $\frac{2R}{5}$ (3) $\frac{3R}{5}$ (4) $\frac{R}{3}$

42. In the figure a part of circuit is shown:



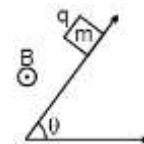
- (1) current will flow from A to B
 (2) current may flow from A to B
 (3) current will flow from B to A
 (4) the direction of current will depend on r

43. Eight wires cut the page perpendicularly at the points shown in figure. A wire labeled with the integer k ($k = 1, 2, \dots, 8$) bears the current $k i_0$. For those with odd k , the current flows up out of the page; for those with even k it flows down into the page. The value of $\oint \vec{B} \cdot d\vec{l}$ along the close path (as shown in the figure) in the direction indicated by the arrow is –



- (1) $10\mu_0 i_0$ (2) $-10\mu_0 i_0$
 (3) $-4\mu_0 i_0$ (4) $4\mu_0 i_0$

44. A block of mass m & charge q is released on a long smooth inclined plane magnetic field B is constant, uniform, horizontal and parallel to surface as shown. Find the time from start when block loses contact with the surface.



- (1) $\frac{m \cos \theta}{qB}$ (2) $\frac{m \operatorname{cosec} \theta}{qB}$
 (3) $\frac{m \cot \theta}{qB}$ (4) None of these

45. A rectangular loop carrying a current I is placed in a uniform magnetic field B . The area enclosed by the loop is A . If there are n turns in the loop, the torque acting on the loop is given by –

- (1) $n i \vec{A} \times \vec{B}$ (2) $n i \vec{A} \cdot \vec{B}$
 (3) $\frac{1}{n} (i \vec{A} \times \vec{B})$ (4) $\frac{1}{n} (i \vec{A} \cdot \vec{B})$

CHEMISTRY

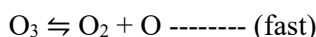
46. For a reaction $pA + qB \rightarrow \text{products}$, the rate law expression is $r = k[A]^l[B]^m$, then :
- $(p + q) < (l + m)$
 - $(p + q) > (l + m)$
 - $(p + q)$ may or may not be equal to $(l + m)$
 - $(p + q) = (l + m)$
47. For the reaction $4A + B \rightarrow 2C + 2D$
The statement which is not correct is :-
- The rate of disappearance of B is one fourth the rate of disappearance of A
 - The rate of appearance of C is half the rate of disappearance of B
 - The rate of formation of D is half the rate of consumption of A
 - The rates of formation of C and D are equal
48. The specific rate constant of a first order reaction depends on the :-
- Concentration of the reactant
 - Concentration of the product
 - Time
 - Temperature
49. For the reaction $A + B \rightarrow \text{products}$, it is observed that :
- (A) on doubling the initial concentration of A only, the rate of reaction is also doubled and
- (B) on doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction.
- The rate of this reaction is given by :
- rate = $k[A][B]$
 - rate = $k[A]^2[B]$
 - rate = $k[A][B]^2$
 - rate = $k[A]^2[B]^2$
50. For an endothermic reaction, energy of activation is E_a and enthalpy of reaction is ΔH (both of these in kJ/mol). Minimum value of E_a will be :-
- Equal to zero
 - Less than ΔH
 - Equal to ΔH
 - More than ΔH
51. The basic strength of the hydrides of group 15 elements :
- decreases on moving down the group
 - increases on moving down the group
 - first decreases upto AsH_3 and then increases
 - first increases upto AsH_3 and then decreases
52. NO_2 can be prepared by heating :
- NH_4NO_3
 - $NaNO_3$
 - $Pb(NO_3)_2$
 - KNO_3
53. Which of the following acids can form two types of salts?
- Hyponitrous acid
 - Nitrous acid
 - Nitric acid
 - Pernitric acid
54. Which of the following statements are correct ?
- Red phosphorus is obtained by heating white phosphorus at 573 K in an inert atmosphere for several days.
 - α -black phosphorus is formed when red phosphorus is heated in a sealed tube at 803 K.
 - β -black phosphorus is obtained by heated white phosphorus at 473 K under high pressure.
- (i) and (ii) only
 - (ii) and (iii) only
 - (i), (ii) and (iii)
 - (i) and (iii) only
55. Which statement is not correct for nitrogen?
- It is obtained by heating $(NH_4)_2Cr_2O_7$
 - It does not readily react with O_2
 - It is a typical non-metal
 - d-orbitals are available for bonding
56. A pungent smelling gas X after being dried by concentrated H_2SO_4 was dissolved in water to give strongly acidic solution. The gas also gives dense white fumes with NH_3 . X is also a constituent of aqua-regia. Which of the following is false for X?
- X is HCl
 - X is Cl_2
 - X is the most volatile among the hydrides of halogens
 - Solution of X in water can liberate CO_2 from the solution of sodium hydrogencarbonates.
57. Among the following which reactions is impossible.
- $F_2 + H_2O \rightarrow HF + O_2$
 - $Cl_2 + H_2O \rightarrow HCl + HClO$
 - $Br_2 + H_2O \rightarrow HBr + HBrO$
 - $I_2 + H_2O \rightarrow HI + HIO$
58. Which of the following interhalogen compounds is impossible :
- ClF_3
 - IF_5
 - $FeCl_3$
 - BrF_5
59. Which of the following options is false for XeF_6 ?
- Its partial hydrolysis gives XeO_3 .
 - Its reaction with silica gives $XeOF_4$
 - It is prepared by the reaction of XeF_4 and O_2F_2 .
 - Its reaction with XeO_3 gives $XeOF_4$.
60. Identify the incorrect statement among the following.
- Cl_2 reacts with excess of NH_3 to give N_2 and HCl.
 - Br_2 reacts with hot and strong NaOH solution to give NaBr, $NaBrO_4$ and H_2O .
 - Ozone reacts with SO_2 to give SO_3 .
 - Silicon reacts with NaOH(aq) in the presence of air to give Na_2SiO_3 and H_2O .

61. **Assertion :** Temperature coefficient of an one step reaction may be negative.

Reason : The rate of reaction having negative order with respect to a reactant decreases with the increase in concentration of the reactant.

- (1) Assertion is true and reason is correct explanation for assertion.
 (2) Assertion is true and reason is false.
 (3) Assertion and reason both true but reason is not correct explanation for assertion.
 (4) Assertion is false and reason is true.

62. The chemical reaction $2\text{O}_3 \longrightarrow 3\text{O}_2$ Proceeds as to follows



The rate law expression should be –

- (1) $r = K [\text{O}_3]^2$
 (2) $r = K [\text{O}_3]^2 [\text{O}_2]^{-1}$
 (3) $r = K [\text{O}_3] [\text{O}_2]$
 (4) Unpredictable
63. The fluoride which is most soluble in water is
 (1) CaF_2 (2) BaF_2 (3) SrF_2 (4) BeF_2

64. The best reducing agent among the following is :-

- (1) NH_3 (2) SbH_3
 (3) PH_3 (4) AsH_3

65. The following data pertain to reaction between A and B :

S. No.	[A] (mol.L ⁻¹)	[B] (mol.L ⁻¹)	Rate (mol.L ⁻¹ t ⁻¹)
I	1×10^{-2}	2×10^{-2}	2×10^{-4}
II	2×10^{-2}	2×10^{-2}	4×10^{-4}
III	2×10^{-2}	4×10^{-2}	4×10^{-4}

Which of the following inference(s) can be drawn from the above data :-

- [a] Rate constant of the reaction is 10^{-4} s^{-1}
 [b] Rate law of the reaction is $k [\text{A}]^1 [\text{B}]^0$
 [c] Rate of reaction increases four times on doubling the concentration of both the reactants

Select the correct answer :-

Codes :

- (1) a, b and c (2) a and b
 (3) b and c (4) b alone
66. Differential form of the rate equation is :

$$\frac{dx}{dt} = k[\text{P}][\text{Q}]^{0.5}[\text{R}]^{0.5}$$

which statement about the above equation is wrong ?

- (1) Order with respect to P is one
 (2) Total order of the reaction is two
 (3) Order with respect to each of Q and R is 0.5
 (4) Unit of specific reaction rate is mol s^{-1}

67. For a gaseous reaction the rate $\left(\frac{dC}{dt}\right)$ is equal to (where C is concentration and n is no. of moles):-

$$(1) \frac{dC}{dt} = \frac{1}{V} \left(\frac{dn}{dt}\right) = \frac{1}{RT} \left(\frac{dP}{dt}\right)$$

$$(2) \frac{dC}{dt} = \left(\frac{dn}{dt}\right) = \frac{dP}{dt}$$

$$(3) \frac{dC}{dt} = \left(\frac{dn}{dt}\right) = \frac{V}{RT} \left(\frac{dP}{dt}\right)$$

- (4) None of them

68. A gaseous reaction $\text{A}_{2(g)} \rightarrow \text{B}_{(g)} + \frac{1}{2}\text{C}_{(g)}$ shows

increase in pressure from 100 mm of Hg to 120 mm of Hg in 5 minutes. What will be rate of disappearance of A_2 (in mm of Hg/min) ?

- (1) 8 (2) 4 (3) 40 (4) 10

69. The rate constant for a first order decomposition reaction is given by

$\log K = 10 - \frac{1000}{T}$. Then, what will be activation energy in kcal/mol ?

- (1) 4.60 (2) 17.6 (3) 3.2 (4) 9.8

70. Azo isopropane decomposes according to the equation

$(\text{CH}_3)_2\text{CHN}=\text{NCH}(\text{CH}_3)_2(g) \rightarrow \text{N}_2(g) + \text{C}_6\text{H}_{14}(g)$. It is found to be a first order reaction. If initial pressure is P_0 and pressure of the reaction mixture at time t is P_t then rate constant K would be:

$$(1) K = \frac{2.303}{t} \log \frac{P_0}{2P_0 - P_t}$$

$$(2) K = \frac{2.303}{t} \log \frac{P_0 - P_t}{P_0}$$

$$(3) K = \frac{2.303}{t} \log \frac{P_0}{P_0 - P_t}$$

$$(4) K = \frac{2.303}{t} \log \frac{2P_0}{2P_0 - P_t}$$

71. Which of the following factor can affect activation energy (E_a)?

- (1) Temperature (2) Reaction stoichiometry
 (3) Catalyst (4) Both 2 & 3

72. What is the activation energy for a reaction if its rate doubles when the temperature is raised from.

20°C to 35°C ? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) 15.1 kJ mol^{-1} (2) 342 kJ mol^{-1}
 (3) 269 kJ mol^{-1} (4) 34.7 kJ mol^{-1}

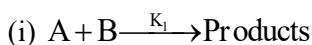
73. The data for the reaction $A + B \rightarrow C$ is

Exp.	[A] ₀	[B] ₀	Initial rate
1	0.012	0.035	0.10
2	0.024	0.070	0.80
3	0.024	0.035	0.10
4	0.012	0.070	0.80

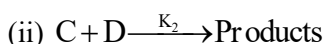
The rate law correspond to the above data is

- (1) Rate = $k[B]^2$
- (2) Rate = $k[B]^4$
- (3) Rate = $k[B]^3$
- (4) Rate = $k[A]^2[B]^2$

74. Given the following second order reactions occurring at a certain temperature (T_1)–



and



The activation energy for reaction (i) is E_{a1} and for(ii) it is E_{a2} such that $E_{a1} > E_{a2}$. If the temperature of both the system is increased from T_1 to T_2 , choose the correct relation from the following–

- (1) $\frac{K_1(T_2)}{K_1(T_1)} = \frac{K_2(T_2)}{K_2(T_1)}$
- (2) $\frac{K_1(T_2)}{K_1(T_1)} > \frac{K_2(T_2)}{K_2(T_1)}$
- (3) $\frac{K_1(T_2)}{K_1(T_1)} < \frac{K_2(T_2)}{K_2(T_1)}$
- (4) None of the above

75. The rate constant of a reaction is given by $K = 2.1 \times 10^{10} \exp(-2700/RT)$ it means that:-

- (1) $\log K$ vs $\frac{1}{T}$ will be a straight line with slope = $-\frac{2700}{2.303R}$
- (2) $\log K$ vs $\frac{1}{T}$ will be a straight line with intercept on $\log K$ axis = $\log 2.1 \times 10^{10}$
- (3) The number of effective collisions are $2.1 \times 10^{10} \text{ cm}^{-3} \text{ sec}^{-1}$
- (4) Both (1) & (2)

76. Match the columns:

Column-I		Column-II	
(A)	Number of collisions per second per unit volume of the reaction mixture.	(p)	Effective collisions.
(B)	Fraction of molecules with energies equal to or greater than E_a	(q)	Collision frequency
(C)	Molecules for which Rate = $Z_{AB} e^{-E_a/RT}$ shows significant deviations	(r)	$e^{-E_a/RT}$
(D)	Collision in which molecules collide with sufficient K.E. and proper orientation	(s)	Complex molecules

- (1) A - (r), B - (p), C - (s), D - (q)
- (2) A - (r), B - (s), C - (q), D - (p)
- (3) A - (q), B - (s), C - (p), D - (r)
- (4) A - (q), B - (r), C - (s), D - (p)

77. Which of the following statements are correct-

- (A) The decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at high pressure.
- (B) Molecules should collide with a sufficient threshold energy and proper orientation for the collision to be effective.
- (C) Molecularity is applicable to elementary as well as complex reaction.
- (D) In complex reaction molecularity and order of the slowest step is same.

Choose correct answer from the options given below -

- (1) A, B and C
- (2) A, B and D
- (3) B and D
- (4) B, C and D

78. Match the Column-I with Column-II

Column-I		Column-II	
(A)	Zero order reaction	(p)	$t_{1/2} = \frac{1}{[A]_0 K}$
(B)	First order reaction	(q)	$t_{100\%} = [A]_0/k$
(C)	Pseudo first order reaction	(r)	$K = K'$ [excess reactant]
(D)	Second order reaction	(s)	$[A]_t = [A]_0 e^{-kt}$

Choose the correct options

- | | | | | |
|-----|----------|----------|----------|----------|
| | A | B | C | D |
| (1) | q | s | p | r |
| (2) | s | q | r | p |
| (3) | q | s | r | p |
| (4) | p | r | s | q |

- 79. Assertion :** The enthalpy of a reaction remains constant in the presence of a catalyst.
Reason: The catalyst participating in the reaction, forms different activated complex and lowers down the activation energy and it catalyses the forward as well as the backward reactions to the same extent.
 (1) Both assertion and reason are true and reason is a correct explanation of the assertion.
 (2) Both assertion and reason are true but reason is not a correct explanation of the assertion.
 (3) The assertion is true but reason is false.
 (4) Both the assertion and reason are false.
- 80.** The activation energy for the reaction $2\text{HI}_{(g)} \rightarrow \text{H}_{2(g)} + \text{I}_{2(g)}$ is 191.5 kJ/mol at 500 K calculate the fraction of molecules of reactants having energy equal to or greater than activation energy -
 (1) 10^{-18} (2) 10^{-20} (3) 10^{-22} (4) 10^{-24}
- 81.** The solubility of iodine in H_2O may be increased by the addition of
 (1) $\text{Na}_2\text{S}_2\text{O}_3$ (2) CHCl_3
 (3) KI (4) CS_2
- 82.** The property of halogen acids, that is indicated incorrectly -
 (1) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$acidic strength
 (2) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$reducing strength
 (3) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$bond length
 (4) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$ thermal stability
- 83.** Which of the following halogens exhibits only one oxidation state in its compounds?
 (1) Bromine (2) Chlorine
 (3) Fluorine (4) Iodine
- 84.** By which of the following reaction SO_2 is formed?
 (1) Reaction of dil. H_2SO_4 with O_2
 (2) Hydrolysis of dil. H_2SO_4
 (3) Reaction of conc. H_2SO_4 with Cu
 (4) None of these
- 85.** Oxygen and sulphur both are the members of the same group in periodic table but H_2O is liquid while H_2S is gas because
 (1) molecular weight of water is more
 (2) electronegativity of sulphur is more
 (3) H_2S is weak acid
 (4) water molecules are having hydrogen bonds between them
- 86.** Which of the noble gas has highest polarisability?
 (1) He (2) Ar
 (3) Kr (4) Xe
- 87.** Which one of the following orders is not in accordance with the property stated against it?
 (1) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$: Acidic property in water
 (2) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Electronegativity
 (3) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Bond dissociation energy
 (4) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Oxidising power
- 88.** The acidic character of dioxides of members of oxygen family decreases in the order
 (1) $\text{SeO}_2 > \text{SO}_2 > \text{TeO}_2 > \text{PoO}_2$
 (2) $\text{SO}_2 > \text{SeO}_2 > \text{TeO}_2 > \text{PoO}_2$
 (3) $\text{PoO}_2 > \text{TeO}_2 > \text{SeO}_2 > \text{SO}_2$
 (4) $\text{TeO}_2 > \text{PoO}_2 > \text{SeO}_2 > \text{SO}_2$
- 89.** Which of the following hydrides is most acidic?
 (1) H_2Te (2) H_2Se
 (3) H_2O (4) H_2S
- 90.** Which one of the following does not undergo hydrolysis?
 (1) AsCl_3 (2) SbCl_3
 (3) PCl_3 (4) NF_3

BIOLOGY

91. The female children of haemophilic father and carrier mother are
 (1) All haemophilic
 (2) Half haemophilic, half carrier
 (3) All normal
 (4) All carrier
92. In which of the following, there is no defect in the sex chromosome?
 (1) Turner's syndrome
 (2) Down's syndrome
 (3) Colour blindness
 (4) Klinefelter's syndrome
93. Find out correct RNA transcribed from this segment of DNA.
 3' ATGCATGCA 5'
 5' TACGTACGT 3'
 (1) 3' AUGCAUGCA 5'
 (2) 5' UACGUACGU 3'
 (3) 3' ACGUACGUA 5'
 (4) 5' ACGUACGUA 3'
94. What is the correct sequence of formation of various structures in order of origin of life?
 I. Protobionts
 II. Simple organic molecules
 III. Eobionts
 IV. Complex organic molecules
 (1) I, II, III and IV (2) II, I, III and IV
 (3) II, IV, I and III (4) III, II, I and IV
95. Genetic map is one that:-
 (1) Shows the stages during the cell division
 (2) Shows the distribution of various species in a region
 (3) Establishes sites of the genes on a chromosome
 (4) Establishes the various stages in gene evolution
96. Klinefelter's syndrome results from
 (1) XX egg and Y sperm
 (2) XX egg and XY sperm
 (3) X egg and YY sperm
 (4) XY egg and X sperm
97. Select the most appropriate sentence.
 i. Water vapour, ethane, carbondioxide and ammonia released from molten mass covered the surface.
 ii. The UV rays from the sun brokeup water into Hydrogen and Oxygen and the heavier escaped.
 iii. Oxygen combined with ammonia and methane to form water, and others. The ozone layer was formed.
 iv. As it cooled, the water vapor fell as rain, to fill all the depressions and form oceans.
 (1) Only i is correct
 (2) Only i and ii are correct
 (3) Only ii and iii are correct
 (4) Only iii and iv are correct
98. The most apparent changes during the evolutionary history of *Homo sapiens* is traced in:
 (1) Walking upright
 (2) Remarkable increase in the brain size
 (3) Shortening of jaws
 (4) Loss of body hair
99. In Hardy-Weinberg law the homozygous dominant, homozygous recessive and heterozygous individuals are represented by respectively
 (1) $p^2 + q^2 + 3pq$
 (2) $p^2 + q^2 + 2pq$
 (3) $q^2 + p^2 + 2pq$
 (4) $q^2 + 2pq + p^2$
100. Mutation results in-
 (1) Random change in gene frequency
 (2) Stabilization of allele frequency
 (3) Adaptive change in phenotype
 (4) Stabilisation of selection pressure
101. Evolution of different species in a given area starting from a point and spreading to other geographical areas is known as :
 (1) Migration
 (2) Divergent evolution
 (3) Adaptive radiation
 (4) Natural selection
102. Which of the following is not a concept of Lamarck?
 (1) Use and disuse of organs
 (2) Better fit individuals produces more progeny.
 (3) Inheritance of acquired characters
 (4) If an organ is used constantly it will continuously increase its size

- 103.** Select the correct statement :
- Life originated on earth after 4 billion years of origin of earth
 - Prokaryotic cells are formed after the formation of eukaryotic cells
 - Water and oxygen are two essential compounds for the origin of life through chemosynthesis
 - First non-cellular forms of life would have been giant molecules like RNA, protein, polysaccharides etc.
- 104.** Some statements are given for wings of bat and butterfly :
- They are analogous structure
 - They are homologous structure
 - They are result of divergent evolution
 - They perform different functions
 - They are example of adaptive divergence
 - They indicates common ancestry
- How many statements are correct ?
- One
 - Two
 - Three
 - Four
- 105.** Read the statements A and B and choose the correct option as
- A:** Natural selection can lead to stabilization.
B: More individuals acquire mean character value in this selection.
- If both statements are correct.
 - If both statements are incorrect.
 - If only statement A is correct.
 - If only statement B is correct.
- 106.** Hardy –Weinberg principle explains
- Genetic equilibrium
 - Ideal or non-evolving population
 - Evolutionary forces affecting the equilibrium.
 - All of these
- 107.** Only two statements are correct among the given five statements. Identify them :
- Fitness is the end result of ability to adapt and get selected by nature.
 - Members of a population don't vary in characteristics.
 - Mutation is a continuous source of variations.
 - According to Darwin evolution is a single step process called saltation.
 - Evolution is not a directed process in the sense of determinism.
- A, E
 - A, B
 - C, D
 - D, E
- 108.** Read two statements (a) and (b) carefully and select the correct option regarding them:
- The first life on earth was chemoautotrophic and anaerobic.
 - Protobionts could exhibit simple metabolism and reproduction.
- Statement (a) is correct but statement (b) is incorrect
 - Statement (a) is incorrect but statement (b) is correct
 - Both statements are correct
 - Both statements are incorrect
- 109.** Part of chromatin which is densely packed, stain darkly and is transcriptionally inactive named as
- Euchromatin
 - Chromatosome
 - Heterochromatin
 - Chromosome
- 110.** Which option is not an example of analogous organs?
- Wings of butterfly and birds
 - Eyes of octopus and mammals
 - Heart of vertebrates
 - Potato and sweet potato
- 111.** The organism A and B are example of:
- The image shows two side-by-side illustrations of rats. The one on the left is labeled 'A- Placental rat' and the one on the right is labeled 'B- Marsupial rat'.
- Convergent evolution
 - Divergent evolution
 - Adaptive convergence
 - Both (1) & (3)
- 112.** In the protein synthesis, tRNA carrying the amino acid enters at which site of the ribosome during elongation.
- A-site
 - P-site
 - Anticodon site
 - R-site
- 113.** Basis of DNA fingerprinting
- High degree of polymorphism in sequence
 - Low degree of polymorphism in sequence
 - Intermediate degree of polymorphism in sequence
 - Sequences show no polymorphism
- 114.** Herbicide, pesticides resistant varieties in a much lesser time scale are examples of:
- Natural selection
 - Anthropogenic action
 - Adaptive radiation
 - Homology

Direction(Q.115 to Q.119): Read the **assertion** and **reason** carefully to mark the correct option out of the options given below:

- (A) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
 (B) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
 (C) If the assertion is true but the reason is false.
 (D) If both the assertion and reason are false

115. Assertion: DNA acts as a genetic material in most organisms.

Reason: RNA is genetic material in some bacteria.

- (1) A (2) B (3) C (4) D

116. Assertion: Histones are basic in nature.

Reason: These are rich in the amino acids lysine and arginine.

- (1) A (2) B (3) C (4) D

117. Assertion: In eukaryotes, transcription occurs in nucleus.

Reason: In bacteria, transcription and translation occur in cytoplasm.

- (1) A (2) B (3) C (4) D

118. Assertion: Theory of chemical evolution proposed that life came from pre-existing non-living organic molecules.

Reason: Primitive earth condition led to the production of organic molecules.

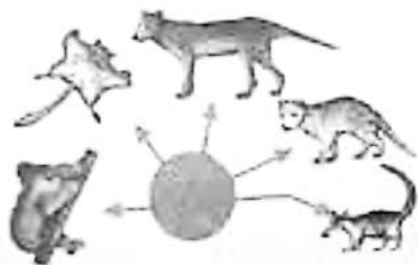
- (1) A (2) B (3) C (4) D

119. Assertion: Hugo de Vries said mutations cause speciation.

Reason: Mutations are slow changes.

- (1) A (2) B (3) C (4) D

120. What is not true about the organisms given below in the diagram?



- (1) They all are example of convergent evolution
 (2) They all are example of adaptive radiation
 (3) They have evolved from a common ancestor
 (4) They all are Australian marsupials

121. Match the following column-I & column-II.

Column-I		Column-II	
A.	Termination (Translation)	i.	Aminoacyl tRNA synthesis
B.	Translation	ii.	Okazaki fragments
C.	Transcription	iii.	GTP dependent release factor
D.	DNA replication	iv.	RNA polymerase

- (1) A-iii, B-i, C-iv, D-ii
 (2) A-iv, B-iii, C-i, D-ii
 (3) A-ii, B-iii, C-i, D-iv
 (4) A-ii, B-i, C-iii, D-iv

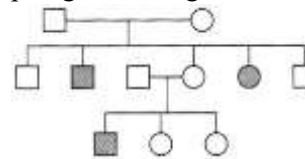
122. A normal women whose father was colourblind, is married to a normal man. The sons would be

- (1) 75% colourblind (2) 50% colourblind
 (3) All normal (4) All colourblind

123. How many are sex linked disorder
 Phenylketonuria, Sickel cell anaemia, Haemophillia, Colour blindness

- (1) One (2) Two (3) Three (4) Four

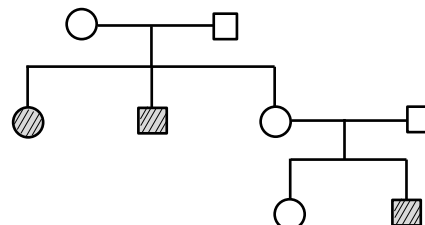
124. Study the pedigree chart given below.



What does it show?

- (1) Inheritance of a sex- linked inborn error of metabolism like phenylketonuria
 (2) Inheritance of a condition like phenylketonuria as an autosomal recessive trait
 (3) The pedigree chart is wrong as this is not possible
 (4) Inheritance of a recessive sex – linked disease like haemophilia

125. Predict from the following chart:–



- (1) Character is x– linked dominant
 (2) Character is x– linked recessive
 (3) Character is Autosomal dominant
 (4) Character is Autosomal recessive

126. Charles Darwin concluded that
- Existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago.
 - Many such life forms do not exist anymore. There had been extinctions of different life forms in the years gone by just as new forms of life arose at different periods of history of earth.
 - There has been fast evolution of life forms.
- (1) i and ii are correct (2) ii and iii are correct
(3) iii is correct (4) All are correct

127. Human females have
- 22 pairs of autosomes and one pair of sex chromosome
 - 21 pairs of autosomes and two pairs of sex chromosome
 - 23 pairs of autosomes and one pair of sex chromosome
 - 20 pairs of autosomes and one pair of sex chromosome

128. A cross in which parents differ in a single pair of contrasting character is called
- Monohybrid cross
 - Dihybrid cross
 - Trihybrid cross
 - Tetra hybrid cross

129. A couple whose all sons are colourblind with AB blood group, identify the parents from the following.
- Mother colourblind with A blood group, and father normal with blood group-B
 - Mother normal with blood group-A, and father colourblind with blood group-B
 - Mother colourblind with blood group-B, and father normal with blood group-B
 - Mother normal with blood group-B, and father colourblind with blood group-A.

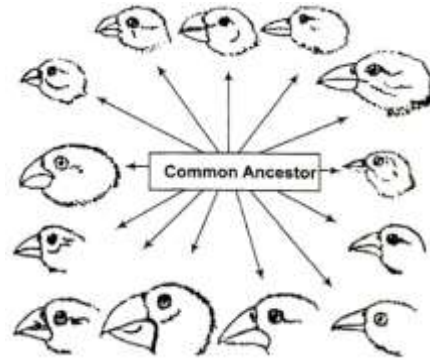
130. Evolution is
- Discontinuous process
 - Continuous process
 - Both (1) and (2)
 - Non-essential process

131. Match column I with column II:
Choose the correct answer from the options given below:

Column I		Column II	
A	Mesozoic Era	i	Lower invertebrates
B	Proterozoic Era	ii	Fish and Amphibians
C	Cenozoic Era	iii	Birds and Reptiles
D	Paleozoic Era	iv	Mammals

- (1) A-i, B-ii, C-iii, D-iv
(2) A-iii, B-ii, C-i, D-iv
(3) A-iv, B-iii, C-ii, D-i
(4) A-iii, B-i, C-iv, D-ii

132. Identify what is indicated in the given diagram?



- (1) Genetic drift
(2) Adaptive radiation
(3) Ecological succession
(4) Different species of finches by mutation

133. Which conclusion can be made on the basis of experiment of Lederberg?

- (1) Preexisting adaptive mutations provide selection advantage to the individuals of a population
(2) Already existed harmful mutations provide selection advantage to the individuals of a population
(3) Individuals of a population develop new characters in response to changed environment which helps in their selection
(4) Any change in environment doesn't affect the individuals of a population

134. Identify the correct sequence of stages in evolution of modern man/ *Homo sapiens*.

- (1) *Australopithecus*, Neanderthal man, Cro-Magnon man, *Homo erectus* and Modern man
(2) *Australopithecus*, *Homo erectus*, Neanderthal man, Cro-Magnon man and Modern man
(3) *Homo erectus*, Neanderthal man, *Australopithecus*, Cro-Magnon man and Modern man
(4) *Homo erectus*, *Australopithecus*, Neanderthal man, Cro-Magnon man and Modern man

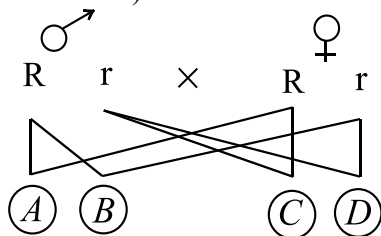
135. According to the theory spontaneous generation of life, originated from

- (1) Non-living matter
(2) Pre-existing life
(3) Chemicals combination
(4) Extra-terrestrial matter

136. Match the following column-I & column-II.

Column-I		Column-II	
A. Lac y gene	i.	Binding site for RNA polymerase	
B. Lac a gene	ii.	Permease enzyme	
C. Promotor	iii.	Repressor binding site	
D. Operator	iv.	Transacetylase enzyme	

- (1) A-ii, B-iv, C-iii, D-i
 (2) A-ii, B-i, C-iv, D-iii
 (3) A-ii, B-iv, C-i, D-iii
 (4) A-iv, B-iii, C-i, D-ii
137. Choose the correct option for the chromosomal disorders
 I. Colour blindness II. Down's syndrome
 III. Phenylketonuria IV. Turner's syndrome
 V. Thalassemia
 (1) I, II and III (2) II, IV and V
 (3) III, IV and V (4) II and IV
138. Fossils in the earth support evolution. Which of the following sentences support the following line?
 (1) A study of fossils in different sedimentary layers indicates the geological period in which the organism existed
 (2) They represent extinct organisms (e.g., Dinosaurs)
 (3) The study showed that life-forms varied over time and certain life forms are restricted to certain geological timespans.
 (4) All of them
139. Genes A, B and C are linked. Genes A and B are more close than A and C. Find out the correct option for the given statements
 I. A might be before B and C
 II. B might be between A and C
 III. C might be between A and B
 IV. More crosses has occurred between A and C than A and B
 (1) I and II (2) II and III
 (3) III and IV (4) I, II and IV
140. Find the phenotype of A, B, C, D from given cross (R- Red and r = white)



- (1) A-Red, B-Red, C-Red, D-White
 (2) A-Red, B-Red, C-White, D-White
 (3) A-Pink, B-Red, C-White, D-White
 (4) A-Pink, B-Red, C-Red, D-White

141. **Statement -I:** Chromosome I has 231 genes and the Y- has 2968 genes.

Statement -II: Repetitive sequences of DNA are thought to have direct coding function and shed light on chromosome structure, dynamic and evolution.

- (1) Both statement I and statement II are true.
 (2) Both statement I and statement II are false.
 (3) Statement I is true but statement II is false.
 (4) Statement I is false but statement II is true.
142. Read (i) to (v) and find the correct option
 i. Nitrogen base is linked to pentose sugar through N-glycosidic linkage
 ii. Phosphate group is linked to 5'-OH of a nucleoside through phosphoester linkage
 iii. Two nucleosides are linked through 3' → 5' N-glycosidic linkage.
 iv. Negatively charged DNA is wrapped around positively charged histone octamer to form nucleosome
 v. Chromatin that is more densely packed and stains dark is called euchromatin.
 (1) i alone is wrong
 (2) iv alone is wrong
 (3) iii and v are wrong
 (4) i, ii, iii wrong
143. The observations about the number of white winged and dark winged moths in England during different environment conditions shows that:
 (1) Predators can't spot a moth against contrasting background
 (2) No variant is completely wiped out
 (3) The areas where industrialization did not occur, the count of melanic moths was High
 (4) Moths that were able to camouflage themselves, didn't survive
144. **Statement A:** Haeckel supported the embryological similarities as an evidence of evolution.
Statement B: embryos of all vertebrates including human develop a row of vestigial gill slits which are absent in adult vertebrates.
 (1) Statement A is correct but Statement B is incorrect.
 (2) Statement A is incorrect but Statement B is correct.
 (3) Both statements are correct.
 (4) Both Statements are incorrect

145. Match the column-I and II:

Column I		Column II	
A	Homo habilis	i	900 cc
B	Homo neanderthalensis	ii	650-800cc
C	Homo erectus	iii	1400 cc
D	Homo sapiens	iv	1350 cc

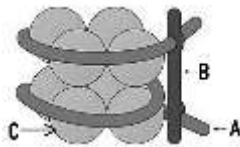
- (1) A – i, B – ii, C – iii, D – iv
 (2) A – ii, B – iii, C – iv, D – i
 (3) A – i, B – ii, C – iv, D – iii
 (4) A – ii, B – iii, C – i, D – iv

146. True statements regarding the genetic drift are

- I. It mostly occurs in smaller population
 II. Certain alleles can be lost forever because of genetic drift
 III. Founder effects and bottle neck effects are caused by genetic drift
 IV. Mutations are primarily responsible for genetic drift

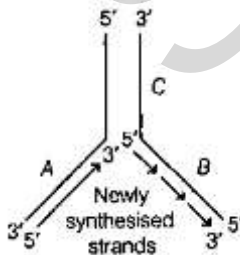
- (1) Only I (2) III and IV
 (3) II and IV (4) All except IV

147. In the given diagram, identify A, B and C.



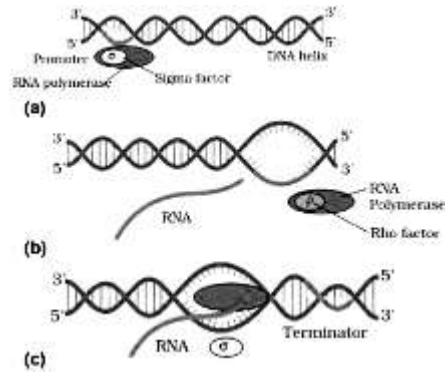
- (1) A–DNA, B–H₁ histone, C–Histone octamer
 (2) A–RNA, B–H₁ histone, C–Histone octamer
 (3) A–DNA, B–H₁ histone, C–Histone tetramer
 (4) A–RNA, B–H₁ histone, C– Histone tetramer

148. Identify A, B and C strands.



- (1) A–Continuous strand, B–Discontinuous strand, C–Template strand
 (2) A–Leading strand, B–Lagging strand, C– Parental strand
 (3) A–5'–3' strand, B–3'–5' strand, C–Parental strand
 (4) None of the above

149.



Identify a, b and c.

- (1) a - Elongation, b - Termination, c - Initiation
 (2) a - Initiation, b - Termination, c - Elongation
 (3) a - Initiation, b - Elongation, c - Termination
 (4) a - Termination, b - Elongation, c – Initiation

150. Which statement is not true about *Australopithecines*?

- (1) They lived approx 2 million years ago
 (2) They probably lived in east African grasslands
 (3) They hunted with stones
 (4) They didn't eat fruit

151. Match the following column-I & column-II.

Column-I		Column-II	
A.	G.J. Mendel	i.	Chromosomal theory of Inheritance
B.	Sturtevant	ii.	Punnett Square
C.	Reginald C. Punnet British geneticist	iii.	Prepared chromosome maps
D.	Sutton & Boveri	iv.	Proposed principles of heredity

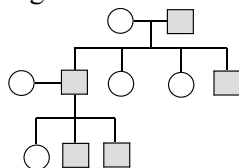
- (1) A-iii, B-iv, C-i, D-ii
 (2) A-i, B-ii, C-iii, D-iv
 (3) A-i, B-iv, C-ii, D-iii
 (4) A-iv, B-iii, C-ii, D-i

152. Which is not correctly matched?

- (1) **Modern man** – First man who started agriculture
 (2) **Neanderthal man** – First man who started to buried dead bodies
 (3) **Dryopithecus** – It was more man like.
 (4) **Cro-Magnon man** –First man who painted walls of caves

153. Mendel published his work on inheritance of characters in 1865, but it remained unrecognized till 1900 due to the reason(s)
- (A) Communication was not easy in those days.
 - (B) His contemporaries did not accept factors as discrete units of heredity.
 - (C) Mathematical approach to explain inheritance was totally new.
 - (D) Mendel could not provide any physical proof for the existence of factors.
- (1) Only A and B are correct
 - (2) Only B and C are correct
 - (3) Only C and D are correct
 - (4) All are correct.

154. Following pedigree chart shows



- (1) Character is sex-linked recessive
 - (2) Character is carried by Y-chromosome
 - (3) Character is sex-linked dominant
 - (4) Character is recessive autosomal
155. Mendel crossed round and yellow seeded pea plant with wrinkled and green seeded plant, then in F_2 generation probability of plants having wrinkled seed would be :
- (1) $\frac{3}{16}$
 - (2) $\frac{1}{16}$
 - (3) $\frac{12}{16}$
 - (4) $\frac{4}{16}$
156. Given below are two statements :
- Statement-I :** In Morgan's experiment, percentage of recombination in 1.3% in case of body colour and eye colour in F_2 in drosophila.
- Statement-II :** Female heterogamety can be seen in hen.
- (1) Both statement-I and statement-II are correct.
 - (2) Statement-I is correct but statement-II is incorrect.
 - (3) Statement-I is incorrect and statement-II is correct.
 - (4) Both statement-I and statement-II are incorrect.
157. Select the correct statements
- I. Natural selection is the major factor for evolution
 - II. Natural selection do not include variations
 - III. Concept of natural selection was given by Darwin
 - IV. Mutation is the sudden inheritable change
 - V. Genetic drift also responsible for evolution.
- (1) I, II and III
 - (2) II, III and IV
 - (3) III, IV and V
 - (4) I, III, IV and V

158. Homology refer to
- I. Divergent evolution
 - II. Common ancestry
 - III. Convergent evolution
- Choose the correct option
- (1) I and III
 - (2) II and III
 - (3) Only III
 - (4) I and II

159. **Assertion:** Unknown DNA after hybridisation with VNTR probe, the autoradiogram gives may bands of differing sizes is DNA profiling.

Reason: These bands gives a characteristic pattern for individual DNA.

- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
 - (3) If assertion is true, but reason is false.
 - (4) If both assertion and reason are false.
160. A dihybrid plant with genotype PpNn. It produce four types of gametes in following number
 $PN = 200$, $pn = 200$, $Pn = 800$, $pN = 800$
 Then what is the distance between linked genes?
- (1) 30 cM
 - (2) 10 cM
 - (3) 25 cM
 - (4) 20 cM
161. Four genes A, B, C and D located on chromosome and cross over frequencies between different genes are :
- $A - D \Rightarrow 20\%$, $A - C \Rightarrow 4\%$
 $C - D \Rightarrow 16\%$, $B - C \Rightarrow 9\%$
 $A - B = 5\%$
- Based on above data, what is the arrangement of linked genes on chromosome.
- (1) A B C D
 - (2) B A C D
 - (3) C B A D
 - (4) C A B D
162. Identify the incorrect option regarding human genome project.
- (1) It was completed in 2003
 - (2) It aims to determine the sequence of 3 billion chemical base pairs and store it in data bases
 - (3) It associated ethical legal and social issues arising from the project
 - (4) It is not associated with non-human organisms DNA sequence.

163. Which one of the following conditions correctly describes the manner of determining the sex in the given example?

- (1) XO type of sex chromosomes determine male sex in grasshopper
- (2) XO condition in humans as found in Turner's syndrome, determines male sex.
- (3) Homozygous sex chromosomes (XX) produce male in *Drosophila*
- (4) Homozygous sex chromosomes, (ZZ) determine female sex in birds.

164. Identify the wrong statement.

- (1) In male grasshoppers, 50% of the sperms have no sex chromosome
- (2) Usually, female birds produce two types of gametes based on sex chromosome
- (3) The human males have one of their sex chromosomes much shorter than other
- (4) In domesticated fowls, the sex of the progeny depends on the type of sperm rather than the egg

165. Match the column and choose the correct

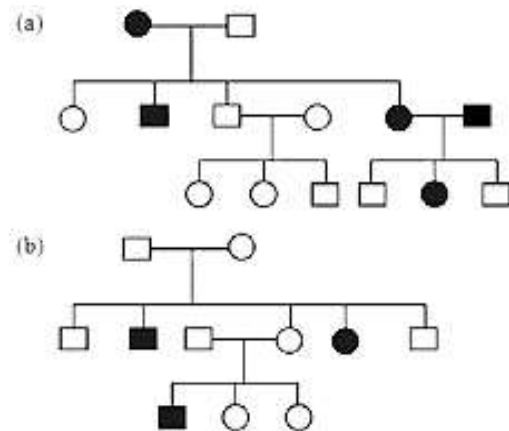
Column-I		Column-II	
A.	Invertebrates formed	i	320 MYA
B.	Sea weeds, few plants formed	ii	350 MYA
C.	Jawless fishes formed	iii	500 MYA
D.	<i>Ichthyosaurus</i> formed	iv	200 MYA

- (1) A-(iii), B-(i), C-(iv), D-(ii)
- (2) A-(i), B-(iii), C-(iv), D-(ii)
- (3) A-(iii), B-(i), C-(ii), D-(iv)
- (4) A-(iv), B-(i), C-(iii), D-(ii)

166. Choose the incorrect pair.

- (1) Promoter – Binding site for RNA polymerase
- (2) Terminator – Define the end of transcription process
- (3) Cistron – Segment of RNA coding for a polypeptide
- (4) Regulatory sequences – Do not code for any RNA or protein

167. Given pedigree (a) & (b) shows the inheritance of a particular genetical disorder. Choose the correct combination.



- (1) a-Sickle cell Anaemia, b-Myotonic dystrophy
- (2) a-Myotonic dystrophy, b-Sickle cell Anaemia
- (3) a-Pseudorickertes, b-colorblindness
- (4) a-colorblindness, b-Pseudorickertes,

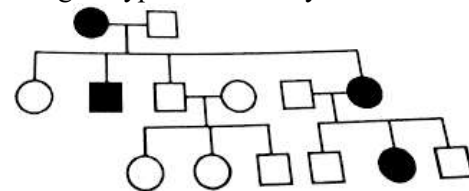
168. Which of the following statement is the most appropriate for sickle-cell anemia?

- (1) It cannot be treated with iron supplements
- (2) It is a qualitative disorder
- (3) It confers resistance to acquiring malaria
- (4) All of the above

169. Which of the following statement describes that natural selection is different from artificial selection

- (1) Natural selection picks the fittest organism, whereas in artificial selection, the breeder decide which organism will breed
- (2) Natural selection depends upon the presence of variation while artificial selection do not
- (3) Natural selection occurs within the population but it is not mandatory in case of artificial selection
- (4) There is a limit of changes that can be brought by natural selection but no such limit exists for artificial selection

170. Given pedigree show inheritance of autosomal dominant disorder (myotonic dystrophy) then what will be genotype of shaded symbol :-



- (1) AA
- (2) Aa
- (3) AA or Aa
- (4) aa

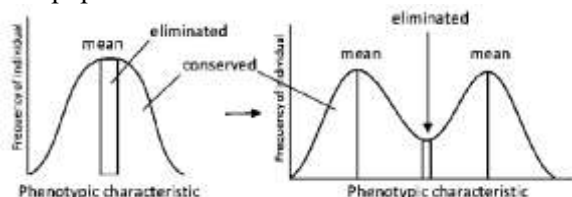
171. Assertion : The possibility of female becoming a haemophilic is extremely rare.

Reason : Mother of haemophilic daughter has to be at least carrier and the father should be haemophilic.

- (1) If both assertion and reason are false
- (2) If assertion is true, but reason is false.
- (3) If both assertion and reason are true and the reason is the correct explanation of the assertion.

- (4) If both assertion and reason are true, but reason is not the correct explanation of the assertion.

172. This graph indicates which type of natural selection in a population?



- (1) Stabilizing selection
- (2) Disruptive selection
- (3) Directional selection
- (4) None of them

173. During DNA fingerprinting, DNA fragments are separated by:

- (1) Gel electrophoresis
- (2) Chromatography
- (3) PAGE
- (4) Centrifugation

Direction: (Q.174 -Q.180) Consider the following statements.

- (A) Statement I is correct, but Statement II is incorrect.
- (B) Statement II is correct, but Statement I is incorrect.
- (C) Both Statement I and Statement II are correct.
- (D) Both Statement I and Statement II are incorrect.

174. Statement I: An organism with lethal mutation may not even develop beyond the zygote stage.

Statement II: All types of gene mutations are lethal.

- (1) A
- (2) B
- (3) C
- (4) D

175. Statement-I : Colour blindness occurs in about 8 percent of female and only about 0.4 percent of males.

Statement-II : The genes that lead to red-green colour blindness are on the X-chromosome.

- (1) A
- (2) B
- (3) C
- (4) D

176. Statement I: Phenyl pyruvic acid is excreted through urine because of its poor absorption.

Statement II: Phenylketonuria is result of single gene mutation.

- (1) A
- (2) B
- (3) C
- (4) D

177. Statement I: In humans, the gamete contributed by the male determines whether the child produced will be male or female.

Statement II: Sex in humans is a polygenic trait depending upon a cumulative effect of some genes on X-chromosome and some on Y-chromosomes.

- (1) A
- (2) B
- (3) C
- (4) D

178. Statement I: Replication and transcription occur in the nucleus but translation occurs in the cytoplasm in eukaryotes.

Statement II: mRNA is transferred from the nucleus into the cytoplasm where ribosomes and amino acids are available for protein synthesis.

- (1) A
- (2) B
- (3) C
- (4) D

179. Statement 1: DNA fingerprinting is very well known for its application in paternity testing is case of disputes.

Statement 2: It employs the principle of polymorphism in DNA sequences as the polymorphisms are inheritable from parent to children

- (1) A
- (2) B
- (3) C
- (4) D

180. Statement I: UAG codon is a termination codon.

Statement II: If in mRNA, a termination codon is present, the protein synthesis stops abruptly whether the protein synthesis is complete or not.

- (1) A
- (2) B
- (3) C
- (4) D

Answer Key and Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	3	2	4	2	2	4	1	3	3	4	4	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	2	4	4	2	3	2	4	1	4	4	3	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	1	2	2	4	3	1	4	4	2	2	3	2	3	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	2	4	3	4	1	3	1	3	4	2	4	3	1	2
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	2	4	2	4	4	1	1	1	1	3	4	3	2	4
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	2	3	1	2	3	1	3	3	4	4	3	2	1	4
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	2	2	3	3	1	4	2	2	1	3	2	4	1	1
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	4	1	4	3	3	4	1	1	2	3	1	2	1	3	1
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	1	2	2	2	4	1	1	1	1	2	4	2	1	2	1
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	3	4	4	4	1	2	3	2	3	4	4	1	1	2	4
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	4	3	4	2	4	1	4	4	2	4	2	4	1	4	3
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	3	2	4	1	2	3	2	1	1	2	3	1	3	3	3

PHYSICS

1. (2)

$$R = \frac{\rho \ell}{A} \quad \ell = \text{resistance per unit length}$$

$$= \frac{\rho}{A} \propto \frac{1}{A}$$

Near A, area of cross-section is less. therefore, resistance per unit length will be more. Hence from the equation, $H = i^2 R t$, heat generation near A will be more.

$$\text{Current density, } J = \frac{i}{A} \text{ or } J \propto \frac{1}{A} \text{ (as } I \text{ is same)}$$

2. (3)

If current flows from a to b, then equation will become

$$V_a - ir - E = V_b \text{ or } V_a = E + ir$$

So, $V_a - V_b$ is always positive. Hence, V_a is always greater than V_b .

3. (3)

$$R = \frac{\ell}{\sigma A}$$

$$\therefore \sigma = \frac{\ell}{RA} \quad \frac{n}{\text{ohm} \cdot \text{m}^2} = \text{ohm}^{-1} \cdot \text{m}^{-1}$$

4. (2)

$$P = \frac{V^2}{R} \text{ or } P \propto \frac{1}{R}$$

$$\frac{P_2}{P_1} = \frac{R_1}{R_2} = \frac{\ell}{\ell} \quad (\text{as } R \propto \ell)$$

$$\therefore P_2 = \left(\frac{\ell}{\ell} \right) \cdot \frac{\ell}{\ell} = 1.11 \text{ m}$$

So, power will increase by 11%.

5. (4)

By symmetry, $V_A = V_B$
or $V_{AB} = 0$

6. (2)

Let $V_0 = V$

Now, $I_{AO} + I_{BO} = I_{OC}$

$$\therefore \frac{6-V}{6} + \frac{3-V}{3} = \frac{V-2}{2}$$

Solving this equation, we get
 $V = 3 \text{ V}$

7. (2)

$$v_d = \frac{i}{neA} = \frac{i}{ne(\pi r^2)} \Rightarrow v_d \propto \frac{i}{r^2}$$

8. (4)

$$V_{20\Omega} = V_{\text{Total}}$$

$$\therefore (20)(0.3) = (R_{\text{Total}})(0.8)$$

$$\therefore \frac{4}{30} = \frac{1}{R_1} + \frac{1}{20} + \frac{1}{15}$$

Solving we get $R_1 = 60\Omega$

9. (1)

Flow of electrons, $\frac{n}{t} = 10^7 / \text{second}$

$$\text{Therefore, current (I)} = \frac{q}{t} = \frac{ne}{t} = \frac{n}{t} \times e$$

$$= 10^7 \times (1.6 \times 10^{-19}) = 1.6 \times 10^{-12} \text{ A}$$

10. (3)

$$R_{ab} = \frac{(3 \times 15)}{3 + 15} = 2.5\Omega$$

$$\text{As } R_{60^\circ} = \left(\frac{18}{360^\circ} \right) (60^\circ) = 3\Omega$$

11. (3)

For $R_3 = 0$, $I = 9 \text{ A}$ (from the graph). In this situation, the circuit can be drawn as shown in figure (A).

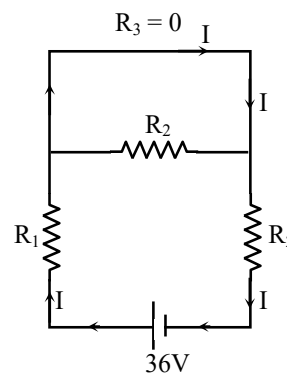


Figure (A)

$$\text{Here } I = \frac{36}{R_1 + R_2} = 9$$

$$\text{or } R_1 + R_2 = 4 \quad \dots(1)$$

For $R_3 \rightarrow \infty$, equivalent resistance of R_2 and R_3 in parallel will be

$$\frac{1}{R_{eq}} = \frac{1}{R_2} + \frac{1}{R_3 \rightarrow \infty} = \frac{1}{R_2}$$

[Current through R_2 will be zero because we have a short circuit ($R_3 = 0$) across R_2]

$$\text{or } R_{eq} = R_2$$

In this situation, the circuit can be drawn as shown figure (B).

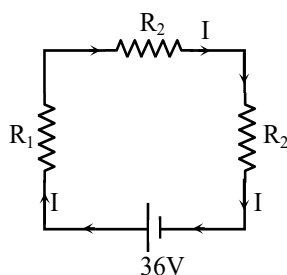


Figure (B)

$$\text{Now } I = \frac{36}{R_1 + 2R_2}$$

$$= 6 \text{ (given)}$$

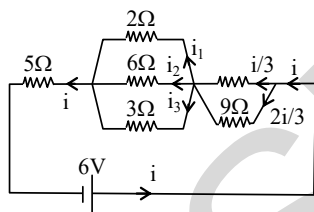
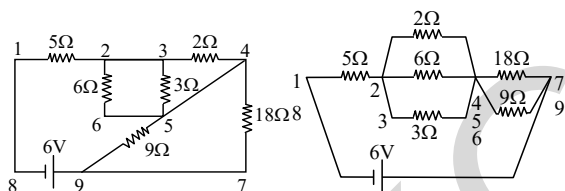
$$\therefore R_1 + 2R_2 = 6 \quad \dots(2)$$

from eqs. (1) and (2)

$$R_1 = 2\Omega, R_2 = 2\Omega$$

12. (4)

Equivalent circuit



$$i_1 : i_2 : i_3 = \frac{1}{2} : \frac{1}{6} : \frac{1}{3} = 3 : 1 : 2$$

$$i_1 = \frac{3}{6} \times i = \frac{i}{2}$$

$$i_2 = \frac{1}{6} \times i = \frac{i}{6}$$

$$i_3 = \frac{2}{6} \times i = \frac{i}{3}$$

$$\Delta H \text{ in } 5\Omega = i^2 \times 5$$

$$\Delta H \text{ in } 2\Omega = \frac{i^2}{4} \times 2 = \frac{i^2}{2}$$

$$\Delta H \text{ in } 5\Omega = \frac{4i^2}{9} \times 9$$

Maximum heat is in 5Ω .

13. (4)

In case of stretching of wire $R \propto l^2$

$\Rightarrow$ If length becomes 3 times so Resistance becomes 9 times i.e. $R' = 9 \times 20 = 180\Omega$

14. (1)

Resistivity does not depend on length & cross section area.

15. (3)

$$i = \frac{q}{t} = \frac{4}{2} = 2 \text{ ampere}$$

16. (2)

As we know that, current is defined as

$$I = \frac{q}{t}$$

Here,

Charge on electron, $q = 1.6 \times 10^{-19} \text{ C}$;

And frequency, $f = \frac{1}{t} = 6.6 \times 10^{15} \text{ Hz}$

$$\therefore I = \frac{q}{t} = qf$$

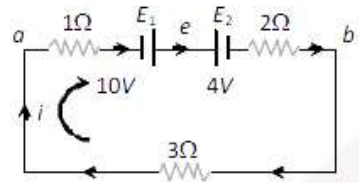
$$\Rightarrow I = 1.6 \times 10^{-19} \times 6.6 \times 10^{15}$$

$$\Rightarrow I = 10.56 \times 10^{-4} \approx 1 \text{ mA}$$

17. (4)

Since $E_1 (10V) > E_2 (4V)$

So current in the circuit will be clockwise.



Applying Kirchhoff's voltage law

$$-1 \times i + 10 - 4 - 2 \times i - 3i = 0$$

$$\Rightarrow i = 1A \text{ (a to b via e)}$$

18. (2)

Let the current in the circuit $= i = \frac{V}{R}$

Across the cell, $E = V + ir$

$$\Rightarrow r = \frac{E - V}{i} = \frac{E - V}{\frac{V}{R}} = \left(\frac{E - V}{V} \right) R$$

19. (4)

In open circuit of a cell $V = E$

20. (4)

Zero (No potential difference across voltmeter)

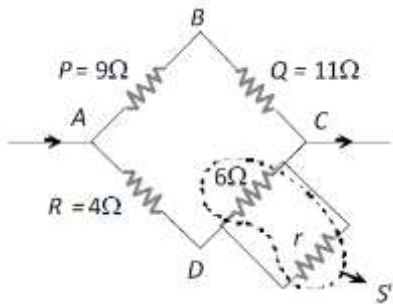
21. (2)

Suppose resistance R is connected in series with voltmeter as shown. Suppose resistance R is connected in series with voltmeter as shown. By Ohm's law.

$$\Rightarrow R = (n - 1) G \left(\text{where } i_g = \frac{V}{G} \right)$$

22. (3)

$$\frac{P}{Q} = \frac{R}{S} \text{ (For balancing bridge)}$$



$$\Rightarrow S' = \frac{4 \times 11}{9} = \frac{44}{9}$$

$$\Rightarrow \frac{1}{S'} = \frac{1}{9} + \frac{1}{6}$$

$$\Rightarrow \frac{9}{44} - \frac{1}{6} = \frac{1}{r}$$

$$\Rightarrow r = \frac{132}{5} = 26.4 \Omega$$

23. (2)

Because for inside the pipe $i = 0$

$$\therefore B = \frac{\mu_0 i}{2\pi r} = 0$$

24. (4)

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{idl \sin \theta}{r^2}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{d\vec{l} \times \vec{r}}{r^3}$$

25. (1)

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \Rightarrow \frac{B_1}{B_2} = \frac{r_2}{r_1} \Rightarrow \frac{10^{-8}}{B_2} = \frac{12}{4}$$

$$\Rightarrow B_2 = 3.33 \times 10^{-9} \text{ Tesla}$$

26. (4)

$$\text{If } \frac{P}{Q} = \frac{20}{80}$$

Now if P & Q are interchanged

$$\frac{Q}{P} = \frac{80}{20} \text{ The balanced point shifts by } 60 \text{ cm.}$$

(4)

Use right hand palm rule, or Maxwell's Cork screw rule or any other.

28. (3)

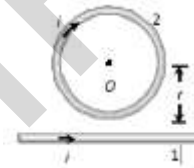
The magnetic field at the centre of the circle

$$= \frac{\mu_0}{4\pi} \times \frac{2\pi i}{r} = 10^{-7} \times \frac{2\pi(nq)}{r}$$

$$= \frac{\mu_0}{4\pi} \times \frac{2\pi i}{r} = 10^{-7} \times \frac{2\pi(nq)}{r} = \frac{2\pi nq}{r} \times 10^{-7}$$

29. (2)

The given shape is equivalent to the following diagram



The field at O due to straight part of conductor is

$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \odot. \text{ The field at } O \text{ due to circular coil}$$

$$\text{is } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \otimes. \text{ Both fields will act in the}$$

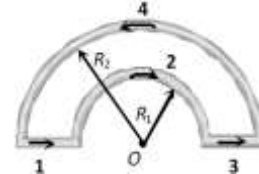
opposite direction, hence the total field at O .

i.e.

$$B = B_2 - B_1 = \left(\frac{\mu_0}{4\pi} \right) \times (\pi - 1) \frac{2i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} (\pi - 1)$$

30. (1)

In the following figure, magnetic fields at O due to sections 1, 2, 3 and 4 are considered as B_1 , B_2 , B_3 and B_4 respectively.



$$B_1 = B_3 = 0$$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{R_1} \otimes$$

$$B_4 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{R_2} \odot \quad \text{As } |B_2| > |B_4|$$

$$\text{So, } B_{\text{net}} = B_2 - B_4$$

$$\Rightarrow B_{\text{net}} = \frac{\mu_0 i}{4} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \otimes$$

31. (1)

Directions of currents in two parts are different, so directions of magnetic fields due to these currents are opposite. Also applying Ohm's law across AB

$$i_1 R_1 = i_2 R_2 \Rightarrow i_1 l_1 = i_2 l_2$$

$$\text{Also, } B_1 = \frac{\mu_0}{4\pi} \times \frac{i_1 l_1}{r^2} \text{ and } B_2 = \frac{\mu_0}{4\pi} \times \frac{i_2 l_2}{r^2}$$

$$(\because l = r\theta)$$

$$\therefore \frac{B_1}{B_2} = \frac{l_1}{l_2} = 1$$

Hence, two field induction's are equal but of opposite direction. So, resultant magnetic induction at the centre is zero and is independent of θ .

32. (1)

33. (3)

Both particles are deflecting in same direction so they must be of same sign. (i.e., both A and B represents protons)

$$\text{By using } r = \frac{mv}{qB} \Rightarrow r \propto v$$

From given figure radius of the path described by particle B is more than that of A. Hence $v_B > v_A$.

34. (2)

$$\text{Because } B = \mu_0 n i \Rightarrow B \propto n i.$$

35. (4)

Since electron is moving parallel to the magnetic field, hence magnetic force on it $F_m = 0$. The only force acting on the electron is electric force which reduces its speed.

36. (3)

Force on portion ab of wire $F_1 = B i l \sin 90^\circ = B i l$
Force on portion bc of wire $F_2 =$

$$B i \left(\frac{l}{\sqrt{2}} \right) \sin 45^\circ = B i l.$$

$$\text{So } \frac{F_1}{F_2} = 1.$$

37. (1)

The gravitational torque must be counter balanced by the magnetic torque about O, for equilibrium of the sphere.

$$\text{The gravitational torque } = \tau_{gr} = m g \times r \sin \theta$$

$$\Rightarrow \tau_{gr} = m g r \sin \theta$$

$$\text{The magnetic torque } \tau_m = \vec{\mu} \times \vec{B}$$

Where the magnetic moment of the coil

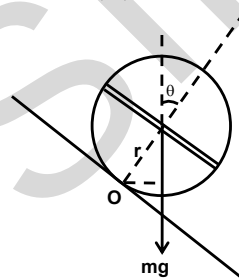
$$= \mu = (i \pi r^2)$$

$$\Rightarrow \tau_m = \pi i r^2 B \sin \theta$$

$$\pi i r^2 B \sin \theta = m g r \sin \theta$$

$$\Rightarrow B = \frac{m g}{\pi i r}$$

Hence (A) is correct.



38. (4)

As the length of wire is doubled, the cross-section area of wire becomes half. Therefore resistance of wire becomes four times and current becomes $\frac{1}{4}$ th of the initial value.

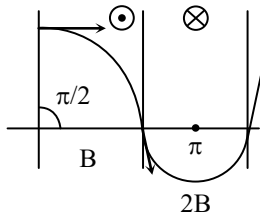
$$\text{Also, } v_d = \frac{I}{n e A}$$

Since current becomes one fourth and cross-section area of wire becomes half, therefore from above equation the drift velocity of electron becomes half. Hence statement I is false.

39. (4)

Statement-1 is obviously false if the current is sent in opposite direction given in figure of statement-2

40. (2)

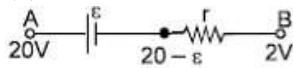


$$\text{Time} = \frac{\pi m}{2qB} + \frac{\pi m}{2qB} = \frac{\pi m}{qB}$$

41. (2)

$$R_{eq} = \frac{R/2 \times 2R}{R/2 + 2R} = \frac{2R}{5}$$

42. (3)



Current through resistance will be from A to B if

$$20 - \varepsilon > \Rightarrow \varepsilon < 18$$

$$\text{And from B to A if } 20 - \varepsilon < 2 \Rightarrow \varepsilon > 18$$

There for current may flow in resistance from B to A

43. (2)

$$\begin{aligned} \bigcirc &= \mu_0(i_1 + i_3 + i_6 + i_8) \text{ (as } i_2, i_4, i_5 \text{ and } i_7, \\ &\text{do not lie inside the loop)} \\ &= \mu_0(1 + 3 - 6 - 8) i_0 = -10 \mu_0 i_0. \end{aligned}$$

44. (3)

Contact loses when $N = 0$

$$V = g \sin \theta \cdot t$$

$$Mg \cos \theta = q g \sin \theta t B \quad [N = 0]$$

$$t = \frac{m \cot \theta}{qB}$$

45. (1)

CHEMISTRY

46. (3)

Sol. Order may be positive negative, or a fractional value.

47. (2)

Sol. $r = \frac{-1}{4} \frac{d[A]}{dt} = \frac{-d[B]}{dt} = \frac{1}{2} \frac{d[C]}{dt} = \frac{1}{2} \frac{d[D]}{dt}$

48. (4)

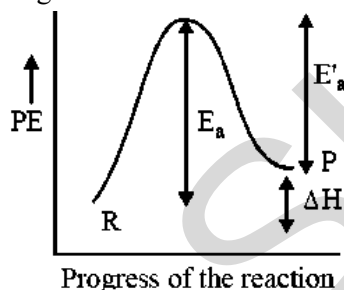
Sol. The rate constant depends only on temperature.

49. (3)

Sol. When concentration of 'A' is doubled, rate is doubled. Hence, order with respect to 'A' is 1. When concentration of both 'A' & 'B' are double, rate increased by 8 times Hence total order is 3.
 $\therefore \text{Rate} = K[A]^1[B]^2$

50. (4)

Sol. In endothermic reactions, energy of reactants is less than that of the products. Potential energy diagram for endothermic reactions is,



where, E_a = activation energy of forward reaction

E'_a = activation energy of backward reaction

ΔH = enthalpy of the reaction

From the above diagram, $E_a = E'_a + \Delta H$. Thus,

$$E_a > \Delta H$$

51. (1)

Sol. The basic strength of the hydrides of group 15 elements down the group decreases with decrease in the electronegativity of the central atom according to Drago's rule.

52. (3)

Sol. $2\text{Pb}(\text{NO}_3)_2 \xrightarrow{673\text{K}} 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$
 NaNO_3 and KNO_3 gives their nitrites and O_2 where as NH_4NO_3 gives N_2O .

53. (1)

Sol. $\text{H}_2\text{N}_2\text{O}_2$ (two replaceable hydrogen) and thus form two series of salts.

54. (3)

Sol. All statements are correct.

55. (4)

Sol. It is 2nd period element which does not contain d-orbitals.

56. (2)

Sol. X is HCl.
 HCl has lowest boiling point used is the most volatile.
 $\text{X} + \text{NaHCO}_3 \longrightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$

57. (4)

Sol. In I_2 opposite reaction of F_2 occurs
 $\text{HI} + \text{O}_2 \longrightarrow \text{I}_2 + \text{H}_2\text{O}$

58. (3)

Sol. FCl_3 is not possible due to unavailability of d-orbitals in fluorine.

59. (1)

Sol. (1) $\text{XeF}_6 + \text{H}_2\text{O} \longrightarrow \text{XeOF}_4 + 2\text{HF}$
 (2) $2\text{XeF}_6 + \text{SiO}_2 \longrightarrow 2\text{XeOF}_4 + \text{SiF}_4$
 (3) $\text{XeF}_4 + \text{O}_2\text{F}_2 \longrightarrow \text{XeF}_6 + \text{O}_2$
 (4) $\text{XeO}_3 + 2\text{XeF}_6 \longrightarrow 3\text{XeOF}_4$

60. (2)

Sol. $3\text{Br}_2 + 6\text{NaOH} \rightarrow 5\text{NaBr} + \text{NaBrO}_3 + 3\text{H}_2\text{O}$
 $\text{O}_3 + \text{SO}_2 \rightarrow \text{O}_2 + \text{SO}_3$
 $\text{Si} + 2\text{NaOH} + \text{O}_2 \rightarrow \text{Na}_2\text{SiO}_3 + \text{H}_2\text{O}$
 Cl_2 reacts with excess of ammonia to produce ammonium chloride and nitrogen.
 $2\text{NH}_3 + 3\text{Cl}_2 \rightarrow \text{N}_2 + 6\text{HCl}$
 $6\text{NH}_3 + 6\text{HCl} \rightarrow 6\text{NH}_4\text{Cl}$
 $\hline 8\text{NH}_3 + 3\text{Cl}_2 \rightarrow \text{N}_2 + 6\text{NH}_4\text{Cl}$

61. (4)

Sol. If order < 0

$$\text{Rate} = \frac{k}{[A]^n}$$

62. (2)

Sol. $\therefore r = K[O][O_3]$

$$K_{eq} = \frac{[O_2][O]}{[O_3]}$$

$$[O] = K_{eq} \frac{[O_3]}{O_2}$$

$$\text{So } r = K K_{eq} \frac{[O_3][O_3]}{[O_2]}$$

$$r = K [O_3]^2 [O_2]^{-1}$$

63. (4)

Sol: Because of hydration energy BeF_2 is most soluble in water

64. (2)

Sol: When an atom loses an electron then element is said to be a reducing agent. The reducing power of an elements depends upon how quickly it can lose electrons.

No. of shells is highest in "Sb" which means the outermost electron is most loosely held in sb. Therefore, SbH_3 is the strongest reducing agent.

65. (4)

Sol. $R = k[A]^1 [B]^0$

$$2 \times 10^{-4} = k[10^{-2}]^1$$

66. (4)

Sol.

67. (1)

Sol.

68. (1)

Sol.

69. (1)

Sol.

70. (1)

Sol.

71. (3)

Sol. Catalyst decreases the value of activation energy

72. (4)

Sol.

73. (3)

Sol. $r = k[A]^m[B]^n$

taking (1) and (4)

$$0.1 = k(0.012)^m (0.035)^n \dots (i)$$

$$0.8 = k(0.012)^m (0.070)^n \dots (ii)$$

(ii)/(i)

$$8 = (2)^n$$

$$n = 3$$

w.r.t. B is third order

taking (1) and (3)

Order w.r.t. A = 0

$$r = k[A]^0[B]^3$$

74. (2)

Sol.

75. (4)

Sol.

76. (4)

Sol.

77. (2)

Sol.

78. (3)

Sol.

79. (1)

Sol. Catalyst catalyse the forward and backward reactions to the same extent but the difference in the energy of reactant and product remains the same.

80. (2)

Sol. $e^{\frac{E_a}{RT}} = x$

$$\frac{-E_a}{RT} \ln e = \ln x$$

$$\frac{-E_a}{RT} = 2.303 \log x$$

$$\frac{-191.5 \times 1000}{8.314 \times 500 \times 2.303} = \log x$$

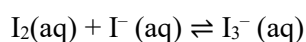
$$-20 = \log x$$

$$x = 10^{-20}$$

81. (3)

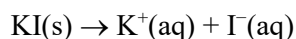
Step 1. Iodine (I_2) is a nonpolar molecule and exhibits very low solubility in polar water due to only weak London dispersion forces.

Step 2. In aqueous solution, iodine can react reversibly with iodide ions to form the triiodide ion:



Step 3. The triiodide ion (I_3^-) is more soluble in water than molecular iodine, so adding iodide shifts the equilibrium right and increases overall iodine solubility.

Step 4. Potassium iodide (KI) dissociates completely in water to provide iodide ions:



Step 5. The supplied iodide ions drive the formation of I_3^- , thus increasing the solubility of I_2 .

Option Analysis:

Option A: $Na_2S_2O_3$ – Thiosulfate chemically reduces I to iodide and oxidizes to tetrathionate, consuming I rather than forming a reversible soluble complex. Incorrect.

Option B: $CHCl_3$ – Chloroform is immiscible with water and dissolves I_2 only in a separate organic layer; it does not enhance aqueous solubility. Incorrect.

Option C: KI – Provides I^- ions that form the soluble triiodide complex; thus it increases iodine solubility. Correct.

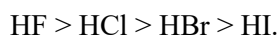
Option D: CS_2 – Carbon disulfide dissolves iodine but is immiscible with water and does not increase iodine solubility in the aqueous phase. Incorrect.

Answer: Option 3 (KI).

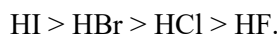
82. (1)

Step 1. Identify the sequences given in each option:

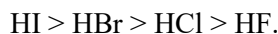
Option A: Acid strength order



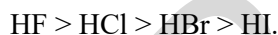
Option B: Reducing strength order



Option C: Bond length order



Option D: Thermal stability order



Step 2. Evaluate acid strength of hydrogen halides. Acid strength increases as bond dissociation energy decreases. Bond strength decreases down the group so acid strength order is $HI > HBr > HCl > HF$.

Thus the sequence in Option A is reversed and incorrect.

Step 3. Evaluate reducing strength. Reducing strength of halide ions increases with ease of oxidation and ionic size, giving $I^- > Br^- > Cl^- > F^-$.

This matches Option B, so it is correct.

Step 4. Evaluate bond length. Bond length increases with atomic size down the group, giving $HI > HBr > HCl > HF$.

This matches Option C, so it is correct.

Step 5. Evaluate thermal stability. Higher bond dissociation energy confers greater thermal stability, so stability order is $HF > HCl > HBr > HI$.

This matches Option D, so it is correct.

Step 6. Conclusion: The only incorrect sequence is in Option 1. Hence, Option 1 is the property indicated incorrectly.

83. (3)

Step 1. List the halogens under consideration: F, Cl, Br, I.

Step 2. Recall that oxidation states depend on the ability to lose or gain electrons, which is governed by electronegativity and availability of vacant orbitals.

Step 3. Note the periodic trend: electronegativity decreases down Group 17. F is the most electronegative and, being in period 2, has no vacant d orbitals to expand its octet, so it cannot exhibit positive oxidation states.

Step 4. Evaluate each option individually:

Option A: Br shows oxidation states $-1, +1, +3, +5, +7$, so it exhibits multiple states.

Option B: Cl shows oxidation states $-1, +1, +3, +5, +7$, so it exhibits multiple states.

Option C: F shows only -1 in all its compounds due to highest electronegativity and lack of d orbitals.

Option D: I shows oxidation states $-1, +1, +3, +5, +7$, so it exhibits multiple states.

Step 5. Thus, the only halogen that exhibits a single oxidation state is F, i.e., Fluorine (Option 3).

84. (3)

Step 1. Evaluate Option A: Reaction of dilute H_2SO_4 with O_2 .

No reaction occurs under normal conditions; sulfuric acid remains in the $+6$ oxidation state and molecular oxygen does not yield SO_2 .

Step 2. Evaluate Option B: Hydrolysis of dilute H_2SO_4 .

Hydrolysis produces H_3O^+ and SO_4^{2-} ions but no sulfur dioxide gas.

Step 3. Evaluate Option C: Reaction of concentrated H_2SO_4 with copper metal (Cu).

Concentrated sulfuric acid acts as a strong oxidizing agent, oxidizing Cu (oxidation state 0) to Cu^{2+} and is itself reduced from sulfur oxidation state $+6$ to $+4$ in SO_2 .

Oxidation states: S in H_2SO_4 is $+6$; in SO_2 is $+4$.

Balanced redox equation:



Step 4. Evaluate Option D: None of these.

This is incorrect since Option C correctly produces SO_2 .

Conclusion: Sulfur dioxide is formed in the reaction of concentrated sulfuric acid with copper (Option 3).

85. (4)

Step One: State the phenomenon.

At 298 K and 1 atm pressure, H_2O is a liquid while H_2S is a gas.

Step Two: Compare molar masses.

Molar mass of H_2O is $2 \times 1.008 + 16.00 = 18.016 \text{ g/mol}$.

Molar mass of H_2S is $2 \times 1.008 + 32.06 = 34.076 \text{ g/mol}$.

If molecular mass alone controlled the boiling point, H_2S (heavier) would have a higher boiling point than H_2O , but experimentally it does not.

Step Three: Examine bond polarity and intermolecular forces.

Electronegativity of O is 3.44 and of S is 2.58. The H–O bond is more polar than the H–S bond, yielding stronger intermolecular attractions in water.

Step Four: Define hydrogen bonding.

A hydrogen bond is a strong dipole–dipole interaction occurring when H is covalently bonded to N, O, or F and is attracted to a lone pair on another electronegative atom.

Step Five: Apply hydrogen bonding to H_2O and H_2S .

Each water molecule can form up to four hydrogen bonds (two as donor, two as acceptor), creating a three-dimensional network. This strong intermolecular force raises its boiling point to 373 K, so it is liquid at 298 K.

Sulphur is less electronegative and larger in size, so H_2S cannot form hydrogen bonds and exhibits only weak van der Waals forces. Its boiling point is 212 K, making it a gas at 298 K.

Step Six: Evaluate the options.

Option A: Incorrect. Molecular weight of water is lower than that of hydrogen sulfide, so mass alone cannot explain the higher boiling point of water.

Option B: Incorrect. Electronegativity of sulphur is lower than that of oxygen, so it cannot account for the higher boiling point of water.

Option C: Incorrect. The acidic strength of H_2S does not determine its physical state at room temperature.

Option D: Correct in principle. Water molecules form hydrogen bonds—a strong intermolecular force—whereas hydrogen sulfide cannot.

Therefore, the principal reason why H_2O is liquid and H_2S is gaseous at room temperature is the presence of hydrogen bonding in water.

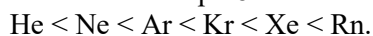
86. (4)

Step 1. List the noble gas atoms: He, Ar, Kr and Xe.

Step 2. Polarizability is the measure of how easily the electron cloud of an atom can be distorted by an external electric field.

Step 3. In noble gases, larger atomic radius and greater number of electrons lead to more loosely held outer electrons, thus higher polarizability.

Step 4. Atomic radius and number of electrons increase down Group 18 in the order



Step 5. Therefore, among the given gases, the polarizability increases as $\text{He} < \text{Ar} < \text{Kr} < \text{Xe}$.

Step 6. Hence, xenon (Xe) has the highest polarizability among the options.

87. (3)

Step 1: List the given orders and their associated properties.

Option & Order and Property

A & $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$: Acidic strength in water

B & $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Electronegativity

C & $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Bond dissociation energy

D & $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Oxidising power

Step 2: Evaluate Option A – Acidic strength of hydrogen halides.

Acid strength in water increases as the H–X bond becomes weaker and the anion better stabilised.

Bond strength decreases down the group: HF (strongest bond) $>$ HCl $>$ HBr $>$ HI (weakest bond). Hence $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ is correct.

Step 3: Evaluate Option B – Electronegativity of halogens.

Electronegativity decreases down a group in the periodic table. Fluorine is the most electronegative, followed by chlorine, bromine, then iodine. Thus $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ is correct.

Step 4: Evaluate Option C – Bond dissociation energy of halogen molecules.

Experimental bond dissociation energies (in kJ/mol) are approximately: $\text{Cl}_2(243) > \text{Br}_2(193) > \text{F}_2(158) > \text{I}_2(151)$. Therefore the correct order is $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$, not $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.

Step 5: Evaluate Option D – Oxidising power of halogens.

Oxidising power correlates with standard reduction potential, which decreases down the group. Hence $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ is correct.

Step 6: Conclusion

Option 3 is not in accordance with the bond dissociation energy trend and is therefore the incorrect order.

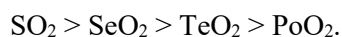
88. (2)

Step 1. Acidic oxides react with water to produce acids, for example $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$.

Step 2. Electronegativity of the central atom decreases down the group from S to Po, decreasing M–O bond polarity.

Step 3. Higher M–O bond polarity increases acid strength, so SO_2 is most acidic and PoO_2 is least.

Step 4. Therefore the acid strength order is



Step 5. Option Analysis:

Option 1 is incorrect because it places SeO_2 above SO_2 .

Option 2 is correct as per the periodic trend of decreasing acidity down the group.

Option 3 is incorrect because it reverses the order completely.

Option 4 is incorrect because it places TeO_2 above SeO_2 .

89. (1)

Step 1. Hydrides of group 16 (oxygen family) include H_2O , H_2S , H_2Se , and H_2Te .

Step 2. Acidity of binary hydrides depends on bond strength (bond enthalpy) and on the ease of proton release to form the conjugate base.

Step 3. Down the group from oxygen to tellurium, the central atom's size increases, weakening the H–E bond and facilitating H^+ release.

Step 4. Bond dissociation enthalpy decreases in the order $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$, so H–E bonds become progressively weaker.

Step 5. Weaker H–E bonds correspond to stronger acidity, giving the acidity order



Step 6. Hence, among the given options, H_2Te is the most acidic hydride.

90. (4)

Step 1. Hydrolysis of covalent halides involves nucleophilic attack by water on the central atom with cleavage of A–Cl bonds, yielding oxoacids and HCl.

Option A: AsCl_3 has polarizable As–Cl bonds and vacant d orbitals, making it prone to hydrolysis.

Reaction: $\text{AsCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{AsO}_3 + 3\text{HCl}$.

Option B: SbCl_3 exhibits similar behavior with polarizable Sb–Cl bonds and vacant d orbitals, undergoing hydrolysis.

Reaction: $\text{SbCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{SbO}_3 + 3\text{HCl}$.

Option C: PCl_3 undergoes hydrolysis forming phosphorous acid due to sufficient bond polarity despite fewer vacant orbitals.

Reaction: $\text{PCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3 + 3\text{HCl}$.

Option D: NF_3 has strong N–F bonds, high bond dissociation energy, negligible vacant orbitals, and low bond polarity, making it resistant to nucleophilic attack by water.

Therefore, NF_3 does not undergo hydrolysis under normal conditions.

Step 2. Conclusion: Among the given options, only NF_3 does not undergo hydrolysis.

Answer: Option 4.